

[Public](#)[Plan Vivo Nature Resources](#)[SOP: Herbaceous Vegetation Surveys](#)

Standard Operating Procedures - Herbaceous Vegetation Surveys

KEY DETAILS

Date : 11 May 2026

Version : 2026.1

Data type : Human Observations

Target groups : Herbaceous Vegetation (grasses, forbs, sedges, and succulents)

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1. PREAMBLE

As a Data Analytics Provider for the Plan Vivo Nature biodiversity credit framework (PVN), Natural State is committed to the highest standards of scientific transparency and integrity. Data collection and processing methods for PVN projects are detailed in a set of Standard Operating Procedures (SOPs). These documents describe the steps involved in all project monitoring methodologies, from data collection to data processing and storage. Each SOP documents key methodological details for a specific data type and target group in accordance with the [PVN Methodology](#).

2. GLOSSARY

Additional definitions are available from Plan Vivo on the [PVN Documents page](#).

Belt transect : A type of elongated vegetation plot (see definition below) that is geographically delineated by a midpoint, two endpoints, and a lateral width. Quadrat locations along a belt transect are used as the sampling units for herbaceous vegetation observations. For the purposes of this SOP, a belt transect is specified as 50 m long and 5 m wide.

Canonical : The accepted scientific binomial of a species, recorded in ODK as `[Genus]_[specific epithet]` (genus and specific epithet joined by an underscore), without authorship.

DAP : Data Analytics Provider. Defined by Plan Vivo as "An independent organisation approved by Plan Vivo to conduct the services of data analysis for the calculation of Plan Vivo Biodiversity Certificates (PVBCs) for Projects in line with the PV Nature Methodology and requirements."

Forb : A herbaceous flowering plant that is not a grass, sedge, or succulent. Forbs are broad-leaved, non-woody species and include most wildflowers and many agricultural weeds.

Grass : A herbaceous plant of the family Poaceae, characterised by narrow leaves, parallel venation, and jointed stems.

Herbaceous Vegetation : Defined for the purposes of this SOP as all live, non-woody plant specimens — comprising grasses, forbs, sedges and succulents — rooted within the bounds of a quadrat at the time of survey.

ODK : ODK (Open Data Kit) is a free, open-source platform that lets organizations build custom digital forms for collecting data in the field — even without an internet connection. It's widely used by researchers, nonprofits, and government agencies to gather structured data. Natural State uses ODK for all field data collection for PVN projects where Natural State serves as the Data Analytics Provider.

Pillar Metric : Defined by Plan Vivo as "A metric that tracks change in one specific aspect of site-level Biodiversity. The Pillar Metrics are aggregated into the Multimetric to produce a single value representing site-level biodiversity change." See the PVN methodology for further details.

Project Area : Defined by Plan Vivo as "A discrete area (terrestrial or aquatic) within which one or more Project Intervention(s) is applied across the entire area and across which biodiversity is

being quantified."

PVBCs : Plan Vivo Biodiversity Certificates. Defined by Plan Vivo as "A certificate issued to a Project by Plan Vivo for a verified Biodiversity Benefit using the PV Nature Methodology." The monetized unit of biodiversity that projects generate and sell to investors.

PVN : Plan Vivo Nature biodiversity framework. A standard for accrediting and certifying projects that restore or conserve biodiversity in accordance with Plan Vivo's core values of community engagement and fair and equitable benefit sharing.

PVN Methodology : Defined by Plan Vivo as "The biodiversity quantification methodology for PV Nature, by which Plan Vivo Biodiversity Certificates (PVBCs) are quantified and quality checked. It sets minimum standards on Biodiversity data quality and sampling protocols, and it defines the Pillar Metrics via which Biodiversity data is aggregated into a Multimetric and then converted into numbers of PVBCs. Under the methodology, projects must collect and report data on a number of broad Target Groups, across a range of trophic levels."

Quadrat : A 1 m X 1 m square sampling frame placed on the ground at at systematic sampling locations within a vegetation plot, within which the presence of all rooted herbaceous species is recorded.

Sedge : A grass-like herbaceous plant of the family Cyperaceae, distinguished from true grasses by typically triangular (three-sided) solid stems, leaves arranged in three vertical ranks, and inconspicuous flowers borne in spikelets. Sedges are common in wet or seasonally waterlogged habitats.

Succulent : A herbaceous or semi-woody plant with thickened, fleshy leaves, stems, or roots adapted to store water, allowing the plant to persist in arid or saline conditions. Common succulent families include Crassulaceae, Aizoaceae, and Euphorbiaceae. For the purposes of this SOP, non-woody succulents rooted within a quadrat are recorded as herbaceous species.

Target Group : Defined by Plan Vivo as "A broad assemblage of species that will be surveyed using the same Data Collection Tool. This is for example, 'fish' rather than a specific type of fish."

3. METHODS OVERVIEW

The survey methods described here document herbaceous species composition within a registered vegetation plot using a fixed series of quadrats placed along the plot's belt transect.

For each quadrat, the field team records the presence of every rooted herbaceous species, captures a top-down photograph of the quadrat, and georeferences the quadrat location. Aggregated across the 20 quadrats sampled per plot, the resulting presence data support the calculation of PVN Pillar Metrics relating to species richness and community composition of the herbaceous vegetation target group.

3.1 METHODOLOGICAL BACKGROUND

These survey methods follow the standard quadrat-based approach to herbaceous vegetation monitoring, in which presence is recorded at the quadrat scale and frequency across quadrats serves as the primary index of relative abundance. It is loosely adapted from the Global Grassy Group's "*Manisa bozaka*" ("Counting grass") protocol (Lehmann et al. 2022: [10.21203/rs.3.pex-1905/v1](#)) but the number and arrangement of quadrats has been adapted to align with Gazelle Ecosolutions' woody survey design (see link in Section 7). The protocol is implemented through the `Survey, Herbaceous Vegetation` ODK form, which enforces a fixed count of 20 quadrats per plot and constrains species selection to a project-specific list of herbaceous species, with a free-text fallback for species not yet on the list. Figure 1 shows the layout of a standard vegetation plot, illustrating the positions of the midpoint, endpoints A and B, and the 5 m belt width relative to the transect midline.

Figure 1. Schematic of a 50 m × 5 m belt transect vegetation plot. The plot is defined by a midpoint and two endpoints (A and B), each 25 m away from the midpoint. The 5 m width (2.5 m on each side of the tape measure) is measured perpendicular to the transect midline.

4. SAMPLING PREPARATION

- The equipment mentioned in the list below needs to be gathered, checked and packed before sampling begins.
- Handheld GPS devices need to be at full battery power and need to be preloaded with the `baseline_sampling_points.gpx` file.
- Mobile devices should be double checked to ensure they are at full battery power and that all necessary ODK forms are loaded.
- Sampling procedures listed here are dependent on vegetation plots first being registered via the `Register, Vegetation Plots` ODK form. This SOP assumes that the plot has been properly laid out in accordance with the PVN Plot Registration SOP.

4.1 EQUIPMENT LIST

- A GPS-enabled Android mobile device or tablet outfitted with a working camera and preloaded with ODK and the relevant project forms.
- A handheld GPS pre-loaded with the survey site point locations. A GPS app on a mobile device can be used. The ODK App should not be used for navigating to sites.
- A rigid quadrat frame of the dimensions specified by the project survey design (typically 1 m × 1 m). A frame constructed from PVC, metal rod, or rope tied around droppers is acceptable provided the inner dimensions are accurate.
- A 50 m (or longer) tape measure used to delineate the 50 m transect.
- A 2.5 m measuring stick or a 3 or 5 m retractable handheld tape measure used to position the quadrats along the edges of the belt transect.
- Extra batteries or charging banks for all electronic devices.
- Herbaceous vegetation identification guides or dichotomous keys appropriate to the project area (for grasses, forbs, sedges and succulents), and the `Pl@ntNet` app loaded on a mobile device separate from the one used for ODK survey forms (recommended).

- Notebook and writing implements (recommended).
- Sample book or sample bags for collecting voucher specimens of unidentified species (recommended).

4.2 DEVICE CONFIGURATION

Ensure that the ODK forms are refreshed before going to the field so that any newly registered vegetation plots appear in the plot picker. It is also recommended to check that the project's herbaceous species list is up to date so that the field team rarely needs to fall back on the form's free-text canonical entry (see Section 5). Confirm the ODK device is set to record geopoints with location accuracy of 5 m or better before leaving for the field.

5. SAMPLING PROCEDURES

1. Ensure plot registration has been completed, using the ODK, **Register, Vegetation Plots** form.
2. If the tape measure delineating the transect is not properly laid out, do so and ensure the tape measure is taut and is not entangled in vegetation. Do not rely solely on stakes or other markers as the tape measure will be used to define the lateral width of the belt transect. Stand at the endpoint A edge of the belt transect. It does not matter which endpoint is A and which is B.
3. Open the **Survey, Herbaceous Vegetation** ODK form and fill in the field team details (mode of transport, all team members involved in the collection activity, and the recorder) and survey details sections. The form will automatically populate the project, project area, replicate, survey type, and target group from the selected survey. Double check that these auto-populated notes correspond to the expected survey details.
4. Select the appropriate vegetation plot on the map in the ODK form. Only plots that have been registered, that belong to the selected survey, and that are recorded as viable will appear. If the plot you are looking for is not present but you expect it to be, try refreshing forms. If it still is not available in the list of registered plots, return to the **Register, Vegetation Plots** form and complete plot registration before continuing. Confirm that the plot status displayed by the form matches what you expect at the site.
5. Starting at endpoint A, place the 2.5 m measuring stick or standard tape measure on the ground with one end touching the transect line and the other end extending 2.5 m to the

- right. Place the quadrat frame flush to the end of the measuring stick/tape measure 1.5 m from the transect line.
6. In the ODK form, indicate whether any herbaceous species are present within the quadrat. A species is considered present if any portion of an individual is rooted within the bounds of the quadrat frame. Foliage that overhangs the quadrat from a plant rooted outside the frame does not count.
 7. If herbaceous species are present, select every species observed from the project's herbaceous species list. If one or more herbaceous species in the quadrat does not appear on the list, indicate this on the new-species prompt and type each missing species as a canonical (Genus_species) in the free-text field, separating multiple canonicals with a single space (e.g., Chloris_virgata Digitaria_eriantha). If a species cannot be identified in the field, give it a distinct provisional name (e.g., Unknown_grass_sp1), record it as you would any other canonical, and collect a voucher specimen and photographs to identify at a later date.
 8. When prompted by the ODK form, hold the mobile device directly above the centre of the quadrat with the camera facing straight down and take a photograph of the quadrat. The entire quadrat frame must be visible in the field of view but try to take the photo as close to ground level as possible without cropping out any of the quadrat. Avoid casting a shadow over the quadrat with your body or the tablet.
 9. Remain standing over the quadrat and record the geopoint when prompted by the form.
 10. If anything about the quadrat is unusual or relevant to interpreting the data (e.g., recent disturbance, presence of dung or termite mounds, standing water, unusually heavy grazing), enter a brief note in the optional quadrat comment field. Otherwise leave the field blank.
 11. Move the 2.5 m measuring stick or tape measure to the left side of the transect and repeat steps 5-11 for the next quadrat.
 12. Quadrat positions are fixed at 5 m intervals along the transect. Walk to the 5 meter mark on the transect tape measure and repeat steps 5-12 at every 5 m interval.
 13. After completing the 20th quadrat, proceed to the finalise-submission section. Review the activity summary, which reports how many of the 20 quadrats had herbaceous species recorded. If these counts look implausibly low (e.g., 0 herbaceous quadrats in a grassland), tap the back button at the bottom of the screen (◀) and review the quadrat data before continuing.
 14. Type into the survey notes field any additional plot-level contextual information not captured elsewhere in the form that is useful for interpreting the data.

15. Confirm the submission. The form will not allow you to advance unless you confirm that the data are complete and correct; if you are uncertain, save the form as a draft instead and revisit it before submitting.
16. Proceed to the next registered veg plot and repeat steps 1–15.

6. POST PROCESSING

Submit any saved forms as soon as a stable internet connection is available. If any unidentified species were recorded under provisional canonicals, identify the voucher specimens and photographs and contact NS to provide proper species identifications at mrogan@naturalstate.org so that both the survey data and the project species lists for subsequent surveys can be updated.

7. OTHER RELEVANT SOPS

[PVN Plot Registration SOP](#)

[PVN Woody Vegetation SOP](#)

8. REVISION AND VERSION HISTORY AND DESCRIPTION

v2026.1 The version of record for the initiation of baseline data collection for the Coexisting with Elephants project (May 2026).